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Compounds for prevention or treatment of neurodegenerative disorders

Abstract

The present disclosure provides a compound of Formula (I) or pharmaceutically acceptable salt, stereoisomers thereof, a pharmaceutical composition comprising the compound, and a method to treat or prevent inflammatory and autoimmune diseases, especially neuroinflammation diseases using the compound. ##STR00001##

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a national phase entry of PCT Application No. PCT/IB2021/057682, filed on 20 Aug. 2021, which claims priority to U.S. Patent Application No. 63/068,483, filed on 21 Aug. 2020. The entire disclosure of the application identified in this paragraph is incorporated herein by reference.

FIELD

(1) The present disclosure generally relates to compounds having enzyme inhibitory activity, pharmaceutical compositions comprising the compound, and methods of using the compounds for treating diseases.

BACKGROUND

(2) Receptor Interacting Protein Kinases (RIPKs) are a family of Ser/Thr and Tyr kinases with important roles in inflammation and innate immunity. The kinase activities of RIPK1 and RIPK3 were found to be critical for the activation of necroptotic cell death pathway by multiple stimuli, including TNF α family of cytokines, interferons (IFNs) and Toll-like receptor (TLR) ligands (Christofferson and Yuan, 2010; Vanlangenakker et al . . . 2012). Importantly, RIPK1/3 kinases have been implicated in a variety of pathologic settings that currently lack effective therapies, including stroke, myocardial infarction, retinal injuries, lethal Systemic Inflammatory Response Syndrome (SIRS) and chronic gut and skin inflammation, and acute pancreatitis (Linkermann and Green, 2014).

(3) RIPK2 is another RIP kinase involved in the activation of NF- κ B, mitogen-activated protein kinases (MAPKs) and apoptosis. The kinase activity of RIPK2 is dispensable for signaling via NF- κ B and the MAPK JNK ('June N-terminal kinase') but is required for activation of the MAPK ERK2. The best-characterized function of RIPK2 is its mediation of signal transduction from the NOD ('nucleotide-binding oligomerization domain') proteins NOD1 and NOD2, which are cytosolic pathogen-recognition receptors that activate pro-inflammatory and antimicrobial responses in response to bacterial peptidoglycans in macrophages. NOD1 and NOD2 are homologous proteins composed of caspase recruitment domains (CARDs), nucleotide-binding domains and leucine-rich repeats. After recognition of its ligands, NOD1 or NOD2 recruits RIPK2 via CARD-CARD homotypic interactions. This process promotes the ubiquitination of RIPK2 and activation of the TAK1 and IKK complexes, which leads to the activation of NF- κ B and MAPKs, as well as the production of pro-inflammatory cytokines by macrophages. RIPK2 also recruits various ubiquitin E3 ligases to the NOD2 complex, including XIAP ('X-chromosome-linked inhibitor of apoptosis'), cIAP1 and cIAP2, PELL3 and LUBAC. It has been demonstrated that XIAP deficiency causes impaired NOD1- or NOD2-mediated ubiquitination of RIPK2 and inflammatory signaling. XIAP is recruited to the NOD2-RIPK2 complex via binding to the kinase domain of RIPK2, which leads to K63-linked ubiquitination of RIPK2 and recruitment of LUBAC; this results in efficient activation of NF- κ B and MAPKs, as well as cytokine production in macrophages. The ubiquitination sites of RIPK2 (Lys209, Lys410 and Lys538) are essential for its function in mediating signaling via NOD1 and NOD2. The D146N kinase-inactive mutant of RIPK2 retains the ability to bind XIAP and to activate NOD2 signaling, which suggests that the kinase activity of RIPK2 is not required for signaling via NOD1 and NOD2.

(4) The NOD2-RIPK2 pathway has attracted special interest due to the role of this signaling node in granulomatous inflammatory diseases, including inflammatory bowel disease (IBD). Such pathologies can arise from either positive or negative dysregulation of the pathway (Caruso et al., 2014; Jostins et al., 2012; Philpott et al., 2014). Genetic variants in NOD2 are the strongest susceptibility factor to Crohn's disease (Hugot et al., 2001; Jostins et al., 2012; Ogura et al., 2001a). Crohn's disease-associated mutations that abrogate NOD2 binding to MDP may induce excessive inflammatory signaling from other pattern recognition receptors, including NOD1 (Couturier-Maillard et al . . . 2013; Inohara et al., 2003). In contrast, mutations in the second major Crohn's disease susceptibility factor, ATG16L1, disrupt an inhibitory interaction with NOD2 and consequently increase the activation of RIPK2 (Sorbara et al., 2013). Excessive RIPK2 activation has also been reported in pediatric Crohn's disease (Negroni et al., 2009). In addition, gain of function in the NOD2-RIPK2 pathway has been linked to Blau syndrome, early-onset sarcoidosis, allergic airway inflammation, and multiple sclerosis (Goh et al., 2013; Jun et al., 2013; Shaw et al., 2011). Overall, these data establish RIPK2 as a key molecule for the understanding of IBD pathogenesis as well as a potential therapeutic target in a wide spectrum of inflammatory and autoimmune diseases, including neuroinflammation diseases.

(5) α -Synuclein is part of a large family of proteins including β - and γ -synuclein and synoretin. α -Synuclein is expressed in the normal state associated with synapses and is believed to play a role in neural plasticity, learning and memory. Several studies have implicated α -synuclein with a central

role in Parkinson disease pathogenesis. Molecular changes in the α -synuclein protein that increase protein misfolding and aggregation have a direct role in disease pathogenesis. Aggregation of α -synuclein contributes to the formation of Lewy bodies and neurites, the pathologic hallmarks of Parkinson disease and α -synucleinopathies. Activation of tyrosine kinase c-abl contributes to α -synuclein-induced neurodegeneration.

(6) The tyrosine kinase c-abl is tightly regulated non-receptor protein tyrosine kinase involved in a wide range of cellular processes, including growth, survival and stress response (Nat Rev Mol Cell Biol, 2004, 5:33-44) and c-abl involved in regulation several cellular processes and has implicated in the development of the central nervous system by controlling neurogenesis. More recently, increasing evidence from various experimental model systems has also revealed that c-abl is activated in neurodegenerative disease such as Alzheimer's disease, Parkinson's disease, Neiman-Pick type C diseases and tauopathies (Human Molecular Genetics, 2014, Vol. 23, No. 11).

(7) The stress-signaling non-receptor tyrosine kinase c-abl links parkin to sporadic forms of Parkinson's disease via tyrosine phosphorylation. Tyrosine phosphorylation of parkin by c-abl is a major post-translational modification that leads to loss of parkin function and disease progression in sporadic Parkinson disease. Inhibition of c-abl offers new therapeutic opportunities for blocking Parkinson disease progression (*The Journal of Neuroscience*. 2011, 31(1): 157-163). Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative disease characterized by progressive death of motor neurons. Knockdown of c-abl with small interfering RNAs (siRNAs) also rescued ALS motor neuron degeneration (Imamura et al., *Sci. Transl. Med.* 9, 2017). Multiple System Atrophy (MSA) is a rare, rapidly progressive neurodegenerative disease without any current treatment. In MSA there is accumulation of α -synuclein in the neurons and oligodendrocytes of the substantia nigra, striatum, olivopontocerebellar structures and spinal cord (*J Neural Transm Vienna Austria* 1996. 2016; 123 (6)).

(8) Administration of the tyrosine kinase inhibitor nilotinib decreases c-abl activity and ameliorates autophagic clearance of α -synuclein in transgenic and lentiviral gene transfer models. Activation of c-abl in the mouse forebrain induces neurodegeneration in the hippocampus and striatum. Therefore, an increase in c-abl activity via phosphorylation may be associated with the α -synuclein pathology detected in Parkinson disease and other neurodegenerative disease (*Hum Mol Genet*. 2013 Aug. 15). c-Abl is a potential therapeutic target for α -synucleinopathy, Parkinson disease, Alzheimer disease, ALS, Dementia with Lewy body and MSA.

(9) Mutations in the leucine-rich repeat kinase 2 (LRRK2) gene are the most common cause of familial Parkinson disease with autosomal dominant inheritance. LRRK2 played important roles in the death of neurons via directly phosphorylating apoptosis signal-regulating kinase 1 at Thr832 site and activating the kinase activity. LRRK2 G2019S mutation impairs dopamine receptor D1 internalization leading to an alteration in signal transduction. Parkinson disease-associated LRRK2 mutations upregulate the expression of mitochondrial calcium uniporter, a mitochondrial calcium transporter and then promote the uptake of dendritic and mitochondrial calcium in cortical neurons and familial Parkinson disease patient fibroblasts (*Frontiers in aging Neuroscience*, 2018 April (10)). Accordingly, LRRK2 has emerged as a promising therapeutic target for disease modification in Parkinson disease.

SUMMARY

(10) The present disclosure provides a compound having RIPK2, c-abl and LRRK2 kinase inhibitory activity, a composition comprising the compound and a method useful to treat inflammatory and autoimmune diseases, especially neuroinflammation diseases. In an embodiment, the compound is a compound of Formula (I):

(11) ##STR00002##

(12) In another embodiment, the present disclosure provides pharmaceutical compositions comprising a therapeutically effective amount of a compound described herein and a pharmaceutically acceptable carrier.

(13) In yet another embodiment, the present disclosure provides methods of inhibiting or treating a neurodegenerative disease comprising administering to a subject in need thereof a therapeutically effective amount of one or more compounds described herein.

Description

DETAILED DESCRIPTION

(1) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

Definitions

(2) The generic terms used in the present disclosure are herein defined for clarity.

(3) This specification uses the terms “substituent”, “radical”, “group”, “moiety”, and “fragment” interchangeably.

(4) As used herein, the term “alkenyl” refers to a straight or branched hydrocarbonyl group with at least one site of unsaturation, i.e., a carbon-carbon, sp² double bond. In an embodiment, alkenyl has from 2 to 12 carbon atoms. In some embodiments, alkenyl is a C₂-C₁₀ alkenyl group or a C₂-C₆ alkenyl group. Examples of alkenyl group include, but are not limited to, ethylene or vinyl (—CH=CH_2), allyl ($\text{—CH}_2\text{CH=CH}_2$), cyclopentenyl ($\text{—C}_5\text{H}_7$), and 5-hexenyl ($\text{—CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH=CH}_2$).

(5) As used herein, the term “alkoxy” is RO— where R is alkyl. Non-limiting examples of alkoxy groups include methoxy, ethoxy and propoxy.

(6) As used herein, the term “alkoxyalkyl” refers to an alkyl moiety substituted with an alkoxy group. Examples of alkoxyalkyl groups include methoxymethyl, methoxyethyl, methoxypropyl and ethoxyethyl.

(7) As used herein, the term “alkoxycarbonyl” is ROC(O)—, where R is an alkyl group as defined herein. In various embodiments, R is a C₁-C₁₀ alkyl group or a C₁-C₆ alkyl group.

(8) As used herein, the term “alkyl” refers to a straight or branched chain hydrocarbonyl group. In an embodiment, alkyl has from 1 to 12 carbon atoms. In some embodiments, alkyl is a C₁-C₁₀ alkyl group or a C₁-C₆ alkyl group. Examples of alkyl groups include, but are not limited to, methyl, ethyl, propyl, isopropyl, butyl, isobutyl, t-butyl, pentyl, hexyl, heptyl, octyl, nonyl and decyl, “lower alkyl” means alkyl having from 1 to 4 carbon atoms.

(9) As used herein, if the term “C₁-C₆” is used, it means the number of carbon atoms is from 1 to 6. For example, C₁-C₆ alkyl means an alkyl which carbon number is any integer of from 1 to 6.

(10) As used herein, the term “alkylamino” refers to an amino group substituted with one or more alkyl groups. “N-(alkyl)amino” is RNH— and “N,N-(alkyl)₂amino” is R₂N—, where the R groups are alkyl as defined herein and are the same or different. In various embodiments, R is a C₁-C₁₀ alkyl group or a C₁-C₆ alkyl group. Examples of alkylamino groups include methylamino, ethylamino, propylamino, butylamino, dimethylamino, diethylamino, and methylethylamino.

(11) As used herein, the term “alkylaminoalkyl” refers to an alkyl moiety substituted with an alkylamino group, wherein alkylamino is as defined herein. Examples of alkylaminoalkyl groups include methylaminomethyl and ethylaminomethyl.

(12) As used herein, the term “alkynyl” refers to a straight or branched carbon-chain group with at least one site of unsaturation, i.e., a carbon-carbon, sp triple bond. In an embodiment, alkynyl has from 2 to 12 carbon atoms. In some embodiments, alkynyl is a C₂-C₁₀ alkynyl group or a C₂-C₆ alkynyl group. Examples of alkynyl groups include acetylenic ($\text{—C}\equiv\text{CH}$) and propargyl ($\text{—CH}_2\text{C}\equiv\text{CH}$).

(13) As used herein, the term “aryl” refers to any monocyclic or bicyclic carbon ring of up to 7 atoms in each ring, wherein at least one ring is aromatic, or an aromatic ring system of 5 to 14 carbon atoms which includes a carbocyclic aromatic group fused with a 5- or 6-membered cycloalkyl group. Representative examples of aryl groups include, but are not limited to, phenyl, tolyl, xylyl, naphthyl, tetrahydronaphthyl, anthracenyl, fluorenyl, indenyl, azulenyl and indanyl. A carbocyclic aromatic group can be unsubstituted or optionally substituted.

(14) As used herein, the term “cycloalkyl” is a hydrocarbyl group containing at least one saturated or partially unsaturated ring structure and attached via a ring carbon. In various embodiments, it refers to a saturated or a partially unsaturated C₃-C₁₂ cyclic moiety, examples of which include cyclopropyl, cyclobutyl, cyclopentyl, cyclopentenyl, cyclohexyl, cyclohexenyl, cycloheptyl and cyclooctyl. “Cycloalkyloxy” is RO—, where R is cycloalkyl.

(15) As used herein, the terms “halogen” and “halo” refers to chloro (—Cl), bromo (—Br), fluoro (—F) or iodo (—I). “Haloalkoxy” refers to an alkoxy group substituted with one or more halo groups and examples of haloalkoxy groups include, but are not limited to, —OCF₃, —OCHF₂ and —OCH₂F. “Haloalkoxyalkyl” refers to an alkyl moiety substituted with a haloalkoxy group, wherein haloalkoxy is as defined herein. Examples of haloalkoxyalkyl groups include trifluoromethoxymethyl, trifluoroethoxymethyl and trifluoromethoxyethyl. “Haloalkyl” refers to an alkyl moiety substituted with one or more halo groups. Examples of haloalkyl groups include —CF₃ and —CHF₂.

(16) As used herein, the term “heteroalkyl” refers to a straight- or branched-chain alkyl group having from 2 to 14 carbons (in some embodiments, 2 to 10 carbons) in the chain, one or more of which has been replaced by a heteroatom selected from S, O, P and N. Exemplary heteroalkyls include alkyl ethers, secondary and tertiary alkyl amines, amides, alkyl sulfides, and the like.

(17) As used herein, the term “heterocyclyl” includes the heteroaryls defined below and refers to a saturated or partially unsaturated monocyclic, bicyclic or tricyclic group of 2 to 14 ring-carbon atoms and, in addition to ring-carbon atoms, 1 to 4 heteroatoms selected from P, N, O and S. In various embodiments the heterocyclic group is attached to another moiety through carbon or through a heteroatom, and is optionally substituted on carbon or a heteroatom. Examples of heterocyclyl include azetidiny, benzoimidazolyl, benzofuranyl, benzofurazanyl, benzopyrazolyl, benzotriazolyl, benzothiophenyl, benzoxazolyl, carbazolyl, carbolinyl, cinnolinyl, furanyl, imidazolyl, indolinyl, indolyl, indolaziny, indazolyl, isobenzofuranyl, isoindolyl, isoquinolyl, isothiazolyl, isoxazolyl, naphthpyridinyl, oxadiazolyl, oxazolyl, oxazoline, isoxazoline, oxetanyl, pyranyl, pyrazinyl, pyrazolyl, pyridazinyl, pyridopyridinyl, pyridazinyl, pyridyl, pyrimidyl, pyrrolyl, quinazolinyl, quinolyl, quinoxalinyl, tetrahydropyranyl, tetrahydrothiopyranyl, tetrahydroisoquinolinyl, tetrazolyl, tetrazolopyridyl, thiadiazolyl, thiazolyl, thienyl, triazolyl, azetidiny, 1,4-dioxanyl, hexahydroazepinyl, piperazinyl, piperidinyl, pyridin-2-onyl, pyrrolidinyl, morpholinyl, thiomorpholinyl, dihydrobenzoimidazolyl, dihydrobenzofuranyl, dihydrobenzothiophenyl, dihydrobenzoxazolyl, dihydrofuranyl, dihydroimidazolyl, dihydroindolyl, dihydroisooxazolyl, dihydroisothiazolyl, dihydrooxadiazolyl, dihydrooxazolyl, dihydropyrazinyl, dihydropyrazolyl, dihydropyridinyl, dihydropyrimidinyl, dihydropyrrolyl, dihydroquinolinyl, dihydrotetrazolyl, dihydrothiadiazolyl, dihydrothiazolyl, dihydrothienyl, dihydrotriazolyl, dihydroazetidiny, methylenedioxybenzoyl, tetrahydrofuranyl, and tetrahydrothienyl, and N-oxides thereof. “Heterocyclyloxy” is RO—, where R is heterocyclyl. “Heterocyclylthio” is RS—, where R is heterocyclyl.

(18) As used herein, the term “3- or 4-membered heterocyclyl” refers to a monocyclic ring having 3 or 4 ring atoms wherein at least one ring atom is heteroatom selected from the group consisting of N, O and S. Non-limiting examples of 3- or 4-membered heterocyclyl include aziridinyl, 2H-aziriny, oxiranyl, thiranyl, azetidiny, 2,3-dihydroazetyl, azetyl, 1,3-diazetidiny, oxetanyl, 2H-oxetyl, thietanyl, and 2H-thietyl.

(19) As used herein, the term “heteroaryl” refers to a monocyclic, bicyclic or tricyclic ring having

up to 7 atoms in each ring, wherein at least one ring is aromatic and contains from 1 to 4 heteroatoms in the ring selected from the group consisting of N, O and S. Non-limiting examples of heteroaryl include pyridyl, thienyl, furanyl, pyrimidyl, imidazolyl, pyranyl, pyrazolyl, thiazolyl, thiadiazolyl, isothiazolyl, oxazolyl, isoxazolyl, pyrrolyl, pyridazinyl, pyrazinyl, quinolinyl, isoquinolinyl, benzofuranyl, dibenzofuranyl, dibenzothiophenyl, benzothienyl, indolyl, benzothiazolyl, benzooxazolyl, benzimidazolyl, isoindolyl, benzotriazolyl, purinyl, thianaphthenyl and pyrazinyl. Attachment of heteroaryl can occur via an aromatic ring, or, if heteroaryl is bicyclic or tricyclic and one of the rings is not aromatic or contains no heteroatoms, through a non-aromatic ring or a ring containing no heteroatoms. "Heteroaryl" is also understood to include the N-oxide derivative of any nitrogen containing heteroaryl. "Heteroaryloxy" is RO—, where R is heteroaryl. (20) As used herein, the term "hydroxyalkoxy" refers to an alkoxy group substituted with a hydroxyl group (—OH), wherein alkoxy is as defined herein. An example of hydroxyalkoxy is hydroxyethoxy.

(21) As used herein, the term "hydroxyalkyl" refers to a linear or branched monovalent C.sub.1-C.sub.10 hydrocarbon group substituted with at least one hydroxy group and examples of hydroxyalkyl groups include, but are not limited to, hydroxymethyl, hydroxyethyl, hydroxypropyl and hydroxybutyl.

(22) As used herein, the term "pharmaceutically acceptable" means suitable for use in pharmaceutical preparations, generally considered as safe for such use, officially approved by a regulatory agency of a national or state government for such use, or being listed in the U.S. Pharmacopoeia or other generally recognized pharmacopoeia for use in animals, and more particularly in humans.

(23) As used herein, the term "pharmaceutically acceptable carrier" refers to a diluent, adjuvant, excipient, or carrier, or other ingredient which is pharmaceutically acceptable and with which a compound of the invention is administered.

(24) As used herein, the term "pharmaceutically acceptable salt" refers to a salt which may enhance desired pharmacological activity. Examples of pharmaceutically acceptable salts include acid addition salts formed with inorganic or organic acids. metal salts and amine salts. Examples of acid addition salts formed with inorganic acids include salts with hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid and phosphoric acid. Examples of acid addition salts formed with organic acids such as acetic acid, propionic acid, hexanoic acid, heptanoic acid, cyclopentanepropionic acid, glycolic acid, pyruvic acid, lactic acid, malonic acid, succinic acid, malic acid, maleic acid, fumaric acid, tartaric acid, citric acid, benzoic acid, o-(4-hydroxy-benzoyl)-benzoic acid, cinnamic acid, mandelic acid, methanesulfonic acid, ethanesulfonic acid, 1,2-ethanedisulfonic acid, 2-hydroxyethane-sulfonic acid. benzenesulfonic acid, p-chlorobenzenesulfonic acid, 2-naphthalenesulfonic acid, p-toluenesulfonic acid, camphorsulfonic acid, 4-methyl-bicyclo[2.2.2]oct-2-ene-1-carboxylic acid, gluco-heptonic acid, 4,4'-methylenebis(3-hydroxy-2-naphthoic) acid, 3-phenylpropionic acid, trimethyl-acetic acid, tertiary butylacetic acid, lauryl sulfuric acid, gluconic acid, glutamic acid, hydroxy-naphthoic acids, salicylic acid, stearic acid and muconic acid. Examples of metal salts include salts with sodium, potassium, calcium, magnesium, aluminum, iron, and zinc ions. Examples of amine salts include salts with ammonia and organic nitrogenous bases strong enough to form salts with carboxylic acids.

(25) As used herein, the term "substituted" means any of above groups (i.e., alkyl, aryl, heteroaryl, heterocycle or cycloalkyl) wherein at least one hydrogen atom of the moiety being substituted is replaced with a substituent. In one embodiment, each carbon atom of the group being substituted is substituted with no more than two substituents. In another embodiment, each carbon atom of the group being substituted is substituted with no more than one substituent. In the case of a keto substituent. two hydrogen atoms are replaced with an oxygen which is attached to the carbon via a double bond. Unless specifically defined, substituents include halogen, hydroxyl, (lower) alkyl, haloalkyl, mono- or di-alkylamino, aryl, heterocycle, —NO₂, B(OH)₂, BPin, —

NR.sub.aR.sub.b, —NR.sub.aC(=O)R.sub.b, —NR.sub.aC(=O)NR.sub.aR.sub.b, —NR.sub.aC(=O)OR.sub.b, —NR.sub.aSO.sub.2R.sub.b, —OR.sub.a, —CN, —C(=O)R.sub.a, —C(=O)OR.sub.a, —C(=O)NR.sub.aR.sub.b, —OC(=O)R.sub.a, —OC(=O)OR.sub.a, —OC(=O)NR.sub.aR.sub.b, —NR.sub.aSO.sub.2R.sub.b, —PO.sub.3R.sub.a, —PO(OR.sub.a)(OR.sub.b), —SO.sub.2R.sub.a, —S(O)R.sub.a, —SO(N)R.sub.a (e.g., sulfoximine), —(R.sub.a)S=NR.sub.b (e.g., sulfilimine) and —SR.sub.a, wherein R.sub.a and R.sub.b are the same or different and independently hydrogen, halogen, amino, alkyl, haloalkyl, aryl or heterocycle, or wherein R.sub.a and R.sub.b taken together with the nitrogen atom to which they are attached form a heterocycle. R.sub.a and R.sub.b may be in the plural based on atoms which those are attached to.

(26) As used herein, the term “therapeutically effective amount” means when applied to a compound of the invention is intended to denote an amount of the compound that is sufficient to ameliorate, palliate, stabilize, reverse, slow or delay the progression of a disorder or disease state, or of a symptom of the disorder or disease. In an embodiment, the method of the present invention provides for administration of combinations of compounds. In such instances, the “therapeutically effective amount” is the amount of a compound of the present invention in the combination sufficient to cause the intended biological effect.

(27) As used herein, the term “treatment” or “treating” as used herein means ameliorating or reversing the progress or severity of a disease or disorder, or ameliorating or reversing one or more symptoms or side effects of such disease or disorder. “Treatment” or “treating”, as used herein, also means to inhibit or block, as in retard, arrest, restrain, impede or obstruct, the progress of a system, condition or state of a disease or disorder. For purposes of this invention, “treatment” or “treating” further means an approach for obtaining beneficial or desired clinical results, where “beneficial or desired clinical results” include, without limitation, alleviation of a symptom, diminishment of the extent of a disorder or disease, stabilized (i.e., not worsening) disease or disorder state, delay or slowing of a disease or disorder state, amelioration or palliation of a disease or disorder state, and remission of a disease or disorder, whether partial or total.

(28) In another embodiment, the compounds of Formula (I) are used for modulating the activity of a RIPK2, c-abl and/or LRRK2 kinase protein.

(29) As used herein, the term “modulating” or “modulation” refers to the alteration of the catalytic activity of a protein kinase. In particular, modulating refers to the activation or inhibition of the catalytic activity of a protein kinase, depending on the concentration of the compound or salt to which the protein kinase is exposed or, more preferably, the inhibition of the catalytic activity of a protein kinase. The term “catalytic activity” as used herein refers to the rate of phosphorylation of tyrosine, serine or threonine under the influence, direct or indirect, of a protein kinase.

(30) The three main classes that pharmacological inhibitors of kinase activity are categorized by are (1) Type I, or “DFG-in” ATP competitive inhibitors, which directly compete with ATP in the ATP binding site (i.e., dual SRC ABL inhibitor dasatinib), (2) Type II, or “DFG-out” ATP competitive inhibitors, which, in addition to binding the ATP binding site also engage an adjacent hydrophobic binding site that is only accessible when the kinase is in an inactivated configuration (i.e., the activation loop is oriented in a conformation that would block substrate binding) (i.e., imatinib, nilotinib), and (3) non-ATP competitive inhibitors that bind at sites outside the ATP binding site that affect the activity of the kinase (i.e., GNF-2).

(31) As used herein, the phrase “compound(s) of this/the disclosure” includes any compound(s) of Formula (I), as well as clathrates, hydrates, solvates, or polymorphs thereof. And, even if the term “compound(s) of the disclosure” does not mention its pharmaceutically acceptable salt, the term includes salts thereof. In one embodiment, the compounds of this disclosure include stereochemically pure compounds, e.g., those substantially free (e.g., greater than 85% ee, greater than 90% ee, greater than 95% ee, greater than 97% ee, or greater than 99% ee) of other stereoisomers. That is, if the compounds of Formula (I) according to the present disclosure or salts thereof are tautomeric isomers and/or stereoisomers (e.g., geometrical isomers and conformational

isomers), such isolated isomers and their mixtures also are included in the scope of this disclosure. If the compounds of the present disclosure or salts thereof have an asymmetric carbon in their structures, their active optical isomers and their racemic mixtures also are included in the scope of this disclosure.

(32) As used herein, the term “polymorph” refers to solid crystalline forms of a compound of this disclosure or complex thereof. Different polymorphs of the same compound can exhibit different physical, chemical and/or spectroscopic properties. Different physical properties include, but are not limited to stability (e.g., to heat or light), compressibility and density (important in formulation and product manufacturing), and dissolution rates (which can affect bioavailability). Differences in stability can result from changes in chemical reactivity (e.g., differential oxidation, such that a dosage form discolors more rapidly when comprised of one polymorph than when comprised of another polymorph) or mechanical characteristics (e.g., tablets crumble on storage as a kinetically favored polymorph converts to thermodynamically more stable polymorph) or both (e.g., tablets of one polymorph are more susceptible to breakdown at high humidity). Different physical properties of polymorphs can affect their processing. For example, one polymorph might be more likely to form solvates or might be more difficult to filter or wash free of impurities than another due to, for example, the shape or size distribution of particles of it.

(33) As used herein, the term “solvate” means a compound or its salt according to this disclosure that further includes a stoichiometric or non-stoichiometric amount of a solvent bound by non-covalent intermolecular forces. Preferred solvents are volatile, non-toxic, and/or acceptable for administration to humans in trace amounts.

(34) As used herein, the term “hydrate” means a compound or its salt according to this disclosure that further includes a stoichiometric or non-stoichiometric amount of water bound by non-covalent intermolecular forces.

(35) As used herein, the term “clathrate” means a compound or its salt in the form of a crystal lattice that contains spaces (e.g., channels) that have a guest molecule (e.g., a solvent or water) trapped within.

Compounds of the Present Disclosure

(36) The present disclosure provides a compound of Formula (I):

(37) A compound of Formula (I):

(38) ##STR00003## or a pharmaceutically acceptable salt thereof, wherein: R.sub.1 is cyclopropyl, cyclobutyl, 3- or 4-membered heterocyclyl or —CF.sub.3, wherein R.sup.1 is optionally substituted with one or more groups selected from the group consisting of halo, alkyl, hydroxyalkyl, haloalkyl, and monoalkylaminoalkyl, R.sub.2 and R.sub.3 are independently —H, halo, C.sub.1-C.sub.3 alkyl, alkoxy, —CF.sub.3, or —OCF.sub.3, R.sub.4, R.sub.5, R.sub.6 and R.sub.7 are independently —H, halo, C.sub.1-C.sub.3 alkyl, alkoxy, aryl, heteroaryl, cycloalkyl, heterocyclyl, heteroalkyl, amino, —SMe, or —CN, wherein R.sub.4, R.sub.5, R.sub.6 and R.sub.7 are optionally substituted with one or more groups selected from the group consisting of halo, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, hydroxyalkyl, trimethylsilylethoxymethyl, —NO.sub.2, —NR.sub.aR.sub.b, —NR.sub.aC(=O)R.sub.b, —NR.sub.aC(=O)NR.sub.aR.sub.b, —NR.sub.aC(=O)OR.sub.b, —OR.sub.a, —CN, —C(=O)R.sub.a, —C(=O)OR.sub.a, —C(=O)NR.sub.aR.sub.b, —OC(=O)R.sub.a, —OC(=O)OR.sub.a, and —OC(=O)NR.sub.aR.sub.b, and R.sub.a and R.sub.b are independently —H, halo, amino, alkyl, or haloalkyl.

(39) In one embodiment, the compound of Formula (I) is selected from compounds according to Formula (II) and pharmaceutically acceptable salts thereof:

(40) ##STR00004## wherein R.sub.8 is selected from the group consisting of —H, halo, C.sub.1-C.sub.3 alkyl, C.sub.1-C.sub.3 hydroxyalkyl and C.sub.1-C.sub.3 haloalkyl, and R.sub.2, R.sub.3, and R.sub.4 are as defined above.

(41) In various embodiments, the compound of Formula (I) selected from the group consisting of

the following compounds and salts thereof: N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropanecarboxamide; (1R,2R)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1R,2S)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2R)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropanecarboxamide; (1S,2R)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1R,2S)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; and (1R,2R)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide.

(42) The compounds of the present disclosure include stereoisomers of the compounds described herein. In some embodiments, the compounds are stereochemically pure compounds such as those substantially free (e.g., greater than 85% ee, greater than 90% ee, greater than 95% ee, greater than 97% ee, or greater than 99% ee) of other stereoisomers. Examples of such stereoisomers include, but are not limited to,

(43) ##STR00005## wherein R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6 and R.sup.7 are as defined above.

(44) In one embodiment, the compound of Formula (I) is selected from compounds according to Formula (III) and pharmaceutically acceptable salts thereof:

(45) ##STR00006## wherein R.sup.8 is selected from the group consisting of —H, halo, C.sub.1-C.sub.3 alkyl, C.sub.1-C.sub.3 hydroxyalkyl, and C.sub.1-C.sub.3 haloalkyl and R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6 and R.sup.7 are as defined above.

(46) Table 1 shows exemplary compounds having the stereochemistry of Formula (III). Examples of the compound of Formula (III) include, but are not limited to, the following compounds and salts thereof: (1S,2S)—N-(5-(1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(6-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)—N-(5-(6-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-methoxy-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(3-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(2-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-amino-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(3-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(3-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-ethyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-(methylthio)-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-isopropyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-cyclopropyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-cyano-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(2-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(3-fluoro-4-methyl-1H-pyrrolo[2,3-b]pyridin-5-

yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2S)—N-(5-(3-chloro-4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(3,4-dichloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; and (1S,2S)—N-(5-(4-chloro-3-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide.

(47) The present disclosure provides pharmaceutically acceptable salts of the compounds described above. The pharmaceutically acceptable salts are as defined above in the definition section. In some embodiments, the salt is hydrochloric acid salt, tartaric acid salt, phosphoric acid salt, or maleic acid salt.

(48) Medical Uses and Methods of Treatment Using the Compounds According to the Present Disclosure

(49) The present disclosure further provides methods for treating a neurodegenerative disease or disorder in a subject having or susceptible to having such a disease or disorder, by administering to the subject a therapeutically effective amount of one or more compounds as described above. In one embodiment, the treatment is preventative treatment. In another embodiment, the treatment is palliative treatment. In another embodiment, the treatment is restorative treatment.

(50) 1. Diseases or Conditions

(51) The compound of the present disclosure for inhibiting RIPK2, c-abl and/or LRRK2 activity is useful for treatment or prevention of a neurodegenerative disease or disorder. The compound can be used for inhibiting or hindering RIPK2, c-abl and/or LRRK2 kinase activity, and for treating a neurodegenerative disease or disorder, or for preventing aggravation of such disease. Thus, the present disclosure provides a method for inhibiting or hindering RIPK2, c-abl and/or LRRK2 activity in a cell, wherein the cell is contacted with an effective amount of a compound of the present disclosure. In one embodiment, such cell is present in a subject (for example, Alzheimer patients). In another embodiment, there is provided a medical use for treating or preventing inflammatory and autoimmune diseases, especially neuroinflammation diseases or disorder in a subject, using the compound according to the present disclosure. The method of the present disclosure comprises administering to a subject in need of treatment or prevention a pharmaceutical composition containing a therapeutically or prophylactically effective amount of RIPK2, c-abl and/or LRRK2 inhibitor. The inflammatory and autoimmune diseases, especially neuroinflammation diseases or disorder includes, but is not limited to, α -synucleinopathy, Parkinson's disease, dementia with Lewy body, multiple system atrophy (MSA), Alzheimer's disease or amyotrophic lateral sclerosis (ALS).

(52) 2. Subjects

(53) Suitable subjects to be treated according to the present disclosure include mammalian subjects. Mammals according to the present disclosure include, but are not limited to, human, canine, feline, bovine, caprine, equine, ovine, porcine, rodents, lagomorphs, primates, and the like, and encompass mammals in utero. Subjects may be of either gender and at any stage of development. In one embodiment, the suitable subject to be treated according to the present disclosure is human.

(54) 3. Administration and Dosing

(55) The compounds of the present disclosure are generally administered in a therapeutically effective amount. The compounds of the present disclosure can be administered by any suitable route in the form of a pharmaceutical composition adapted to such a route, and in a dose effective for the treatment intended. An effective dosage is typically in the range of about 0.001 to about 100 mg per kg body weight per day, preferably about 0.01 to about 50 mg/kg/day, in single or divided doses. Depending on age, species and disease or condition being treated, dosage levels below the lower limit of this range may be suitable. In other cases, still larger doses may be used without harmful side effects. Larger doses may also be divided into several smaller doses, for administration throughout the day. Methods for determining suitable doses are well known in the art to which the present disclosure pertains. For example, Remington: The Science and Practice of

Pharmacy, Mack Publishing Co., 20th ed., 2000 can be used.

(56) Pharmaceutical Compositions, Dosage Forms and Administration Routes

(57) For the treatment of the diseases or conditions referred to above, the compounds described herein or pharmaceutically acceptable salts thereof can be administered as follows:

(58) Oral Administration

(59) The compounds of the present disclosure may be administered orally, including by swallowing, so that the compound enters the gastrointestinal tract, or absorbed into the blood stream directly from the mouth (e.g., buccal or sublingual administration).

(60) Suitable compositions for oral administration include solid, liquid, gel or powder formulations, and have a dosage form such as tablet, lozenge, capsule, granule or powder.

(61) Compositions for oral administration may be formulated as immediate or modified release, including delayed or sustained release, optionally with enteric coating.

(62) Liquid formulations can include solutions, syrups and suspensions, which can be used in soft or hard capsules. Such formulations may include a pharmaceutically acceptable carrier, for example, water, ethanol, polyethylene glycol, cellulose, or an oil.

(63) The formulation may also include one or more emulsifying agents and/or suspending agents.

(64) In a tablet dosage form the amount of drug present may be from about 0.05% to about 95% by weight, more typically from about 2% to about 50% by weight of the dosage form. In addition, tablets may contain a disintegrant, comprising from about 0.5% to about 35% by weight, more typically from about 2% to about 25% of the dosage form. Examples of disintegrants include, but are not limited to, lactose, starch, sodium starch glycolate, croscopovidone, croscarmellose sodium, maltodextrin, or mixtures thereof.

(65) Suitable lubricants, for use in a tablet, may be present in amounts from about 0.1% to about 5% by weight, and include, but are not limited to, talc, silicon dioxide, stearic acid, calcium, zinc or magnesium stearate, sodium stearyl fumarate and the like.

(66) Suitable binders, for use in a tablet, include, but are not limited to, gelatin, polyethylene glycol, sugars, gums, starch, polyvinyl pyrrolidone, hydroxypropyl cellulose, hydroxypropylmethyl cellulose and the like. Suitable diluents, for use in a tablet, include, but are not limited to, mannitol, xylitol, lactose, dextrose, sucrose, sorbitol, microcrystalline cellulose and starch.

(67) Suitable solubilizers, for use in a tablet, may be present in amounts from about 0.1% to about 3% by weight, and include, but are not limited to, polysorbates, sodium lauryl sulfate, sodium dodecyl sulfate, propylene carbonate, diethyleneglycol monoethyl ether, dimethyl isosorbide, polyethylene glycol (natural or hydrogenated) castor oil, HCORTM (Nikkol), oleyl ester, GelucireTM, caprylic/caprylic acid mono/diglyceride, sorbitan fatty acid esters, and Solutol HSTM.

(68) Parenteral Administration

(69) Compounds of the present disclosure may be administered directly into the blood stream, muscle, or internal organs. Suitable means for parenteral administration include intravenous, intramuscular, subcutaneous intraarterial, intraperitoneal, intrathecal, intracranial, and the like. Suitable devices for parenteral administration include injectors (including needle and needle-free injectors) and infusion methods.

(70) Compositions for parenteral administration may be formulated as immediate or modified release, including delayed or sustained release. Most parenteral formulations are aqueous solutions containing excipients, including salts, buffering agents and isotonic agents. Parenteral formulations may also be prepared in a dehydrated form (e.g., by lyophilization) or as sterile non-aqueous solutions. These formulations can be used with a suitable vehicle, such as sterile water. Solubility-enhancing agents may also be used in preparation of parenteral solutions.

(71) Transdermal Administration

(72) Compounds of the present disclosure may be administered topically to the skin or transdermally. Formulations for this topical administration can include lotions, solutions, creams, gels, hydrogels, ointments, foams, implants, patches and the like. Pharmaceutically acceptable

carriers for topical administration formulations can include water, alcohol, mineral oil, glycerin, polyethylene glycol and the like. Topical or transdermal administration can also be performed by electroporation, iontophoresis, phonophoresis and the like.

(73) Compositions for topical administration may be formulated as immediate or modified release, including delayed or sustained release.

(74) Combination Therapy

(75) A pharmaceutical composition according to the present disclosure may contain one or more additional therapeutic agents, for example, to increase the efficacy or decrease the side effects. In some embodiments, accordingly, a pharmaceutical composition further contains one or more additional therapeutic agents selected from active ingredients useful to treat or inhibit diseases mediated directly or indirectly by RIPKs, c-abl and/or LRRK2 kinase. Examples of such active ingredients are, without limitation, agents to treat inflammatory and autoimmune diseases or disorder.

(76) References for Preparing Pharmaceutical Compositions

(77) Methods for preparing pharmaceutical compositions for treating or preventing a disease or condition are well known in the art to which the present disclosure pertains. For example, based on *Handbook of Pharmaceutical Excipients* (7.sup.th ed.), *Remington: The Science and Practice of Pharmacy* (20.sup.th ed.), *Encyclopedia of Pharmaceutical Technology* (3.sup.rd ed.), or *Sustained and Controlled Release Drug Delivery Systems* (1978), pharmaceutically acceptable excipients, carriers, additives and so on can be selected and then mixed with the compounds of the present disclosure for making the pharmaceutical compositions.

(78) The present disclosure provides a compound having various pharmacological effects by inhibiting RIPK2, c-abl and/or LRRK2 activity. a pharmaceutical composition having the compound as an effective agent, a medical use, particularly for treating a neurodegenerative disease or disorder, of the compound, and a method of treatment or prevention comprising administering the compound to a subject in need of such treatment or prevention. The compounds of the present disclosure and pharmaceutically acceptable salts thereof have good safety and high selectivity for RIPK2, c-abl and/or LRRK2, and thus exhibit superior property as a drug.

EXAMPLES

(79) Hereinafter, the present disclosure is described in considerable detail with examples to help those skilled in the art understand the present disclosure. However, the following examples are offered by way of illustration and are not intended to limit the scope of the invention. It is apparent that various changes may be made without departing from the spirit and scope of the invention or sacrificing all of its material advantages.

Synthesis of Formula (I) Compounds

(80) Synthetic method A was used to prepare the compounds of the following. Below, the illustrating synthetic examples of some compounds of the present disclosure are described, and other compounds can be prepared by the similar method to the one described below with different starting or reacting materials.

Synthetic Method A

Example 1. (1S,2S)—N-(5-(1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 hydrochloride

(81) ##STR00007##

Step 1) 2-(4-bromopyridin-2-yl) acetonitrile

(82) To a mixture of (4-bromopyridin-2-yl) methanol (10 g, 53.19 mmol, 1 eq) and TEA (16.15 g, 159.56 mmol, 22.21 mL, 3 eq) in DCM (150 mL) was added MsCl (12.18 g, 106.37 mmol, 8.23 mL, 2 eq) slowly at 0° C. under N.sub.2. The mixture was stirred at 15° C. for 2 h. TLC (dichloromethane:methanol=20:1; R_f=0.56) showed that one main new spot was detected. The mixture was quenched with water (20 mL) at 0° C. and the resulting mixture was diluted with DCM (30 mL). The mixture was separated and the organic phase was washed with brine (20

mL*2), dried over Na.sub.2SO.sub.4, filtered and concentrated at reduced pressure to give 4-bromo-2-(chloromethyl)pyridine (12 g, crude) as yellow oil and this crude product was used directly for the next step. To a mixture of 4-bromo-2-(chloromethyl)pyridine (12 g, 58.12 mmol, 1 eq, crude) and 18-C-6 (1.54 g, 5.81 mmol, 0.1 eq) in CH.sub.3CN (100 mL) was added KCN (7.57 g, 116.24 mmol, 4.98 mL, 2 eq). The mixture was stirred at 15° C. for 16 h. The mixture was concentrated at reduced pressure to give a residue. The residue was dissolved with water (200 mL) and the resulting mixture was extracted with ethyl acetate (50 mL*3). The combined organic phase was washed with brine (50 mL*2), dried over Na.sub.2SO.sub.4, filtered and concentrated at reduced pressure. The residue was purified by silica gel chromatography (1000 mesh silica gel, petroleum ether/ethyl acetate=20/1, 5/1) to give Compound 1 (8.4 g, 42.63 mmol, 73.35% yield) as yellow solid.

(83) .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ =8.46 (d, J=5.4 Hz, 1H), 7.71 (d, J=1.5 Hz, 1H), 7.66 (dd, J=1.8, 5.4 Hz, 1H), 4.23 (s, 2H)

Step 2) 1-amino-4-bromo-2-(cyanomethyl)pyridin-1-ium 4-methylbenzenesulfonate

(84) To a solution of Compound 1 (2.5 g, 12.69 mmol, 1 eq) in DCM (50 mL) was added O-(mesitylsulfonyl) hydroxylamine (6 g, 27.87 mmol, 2.20 eq) which was dissolved with DCM (50 mL) slowly at 0° C. under N.sub.2. The mixture was stirred at 15° C. for 2 h. Lots of solid formed and the mixture was filtered. The filtered cake was collected and washed with DCM (10 mL*2) to give Compound 2 (7.6 g, 18.43 mmol, 72.64% yield) as white solid.

(85) .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.54 (d, J=7.3 Hz, 1H), 7.90 (dd, J=0.6, 2.2 Hz, 1H), 7.88-7.62 (m, 3H), 6.99 (dd, J=2.2, 7.3 Hz, 1H), 6.76 (s, 2H). 6.29 (s, 1H), 2.50 (s, 6H), 2.16 (s, 3H).

Step 3) 5-bromopyrazolo[1,5-a]pyridin-2-amine

(86) A mixture of Compound 2 (1.7 g, 4.12 mmol, 1 eq, four batches) and K.sub.2CO.sub.3 (1.14 g, 8.25 mmol, 2 eq) in MeOH (30 mL) was stirred at 30° C. for 16 h. The mixture was concentrated at reduced pressure to give a residue. The residue was dissolved with ethyl acetate (200 mL) and the resulting mixture was washed with saturated Na.sub.2CO.sub.3 (30 mL*2), brine (30 mL*2), dried over Na.sub.2SO.sub.4, filtered and concentrated at reduced pressure. The residue was purified by silica gel chromatography (300-400 mesh silica gel, Petroleum ether/ethyl acetate=5/1, 1/1) to give Compound 3 (3.1 g, 14.62 mmol, 88.64% yield) as yellow solid.

(87) .sup.1H NMR (400 MHz, CDCl.sub.3) δ 8.00 (d, J=7.2 Hz, 1H), 7.38 (d, J=2.0 Hz, 1H), 6.60 (dd, J=2.0, 7.2 Hz, 1H), 5.70 (s, 1H), 4.04 (s, 2H).

Step 4) (1S,2S)—N-(5-bromopyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide

(88) To a mixture of Compound 3 (3.1 g, 14.62 mmol, 1 eq) and (1S,2S)-2-fluorocyclopropanecarboxylic acid (1.67 g, 16.08 mmol, 1.1 eq) in CH.sub.3CN (40 mL) was added MsCl (3.35 g, 29.24 mmol, 2.26 mL, 2 eq) and 3-methylpyridine (6.81 g, 73.10 mmol, 7.12 mL, 5 eq) slowly at 0° C. under N.sub.2. The mixture was stirred at 30° C. for 3 h. The mixture was concentrated at reduced pressure. The residue was dissolved with ethyl acetate (200 mL) and the resulting mixture was washed with 10% citric acid (30 mL*2), saturated Na.sub.2CO.sub.3 (30 mL*2), brine (30 mL*2), dried over Na.sub.2SO.sub.4, filtered and concentrated at reduced pressure to give a residue. The residue was purified by silica gel chromatography (1000 mesh silica gel, petroleum ether/ethyl acetate=5/1, 3/1) to give Compound 4 (2.3 g, 7.72 mmol, 52.81% yield) as white solid.

Step 5) (1S,2S)-2-fluoro-N-(5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide

(89) Compound 4 (0.345 g, 1.159 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi (1,3,2-dioxaborolane, 0.588 g, 2.318 mmol), [1,1'-bis(diphenylphosphino) ferrocene]dichloropalladium (II) (Pd(dppf)Cl.sub.2, 0.084 g, 0.115 mmol) and potassium acetate (0.398 g, 4.056 mmol) were mixed at the room temperature in DMSO (10 mL) and then stirred at 90° C. for 24 hr, cooled down to the room temperature, filtered through a celite pad to remove solids, and partitioned between

ethyl acetate and water. The organic layer was washed with aqueous saturated sodium chloride solution, separated, dried with anhydrous MgSO_4 , filtered, and concentrated in vacuo. The residue was chromatographed (SiO_2 , 12 g cartridge; ethyl acetate/hexane=0% to 100%) to give Compound 5 as brown solid (0.289 g, 72.3%).

Step 6) (1S,2S)—N-(5-(1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 hydrochloride

(90) To a solution of Compound 5 (150 mg, 434.56 μmol , 1 eq) and 5-bromo-1H-pyrrolo[2,3-b]pyridine (102.75 mg, 521.47 μmol , 1.2 eq) in dioxane (2 mL) and H_2O (0.5 mL) were added Na_2CO_3 (138.18 mg, 1.30 mmol, 3 eq) and $\text{Pd}(\text{dppf})\text{Cl}_2$ (31.80 mg, 43.46 μmol , 0.1 eq) at 15° C. The mixture was stirred at 60° C. for 4 hr. The mixture was poured into petroleum ether (10 mL), filtered with silica gel and the filtrate was concentrated in vacuum. The residue was purified by prep-HPLC (column: Phenomenex Synergi C18 150*25*10 μm ; mobile phase: [water (0.05% HCl)-ACN]; B %: 21%-41%, 9 min), followed by lyophilization. Example 1 (28.2 mg, 68.52 μmol , 15.77% yield, 99.2% purity, 2HCl) was obtained as yellow solid.

(91) ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 12.36 (br s, 1H), 11.16 (s, 1H), 8.76 (d, $J=1.7$ Hz, 1H), 8.69-8.59 (m, 2H), 8.01 (s, 1H), 7.66 (t, $J=2.8$ Hz, 1H), 7.27 (dd, $J=2.0, 7.2$ Hz, 1H), 6.90 (s, 1H), 6.68 (dd, $J=1.5, 3.1$ Hz, 1H), 5.06-4.81 (m, 1H), 2.15 (quin, $J=6.9$ Hz, 1H), 1.73-1.59 (m, 1H), 1.17 (tdd, $J=6.3, 9.0, 12.3$ Hz, 1H).

(92) Table 1 below shows the compounds of Examples along with general synthetic methods used to make the compound and characterization data.

(93) TABLE-US-00001 TABLE 1 Compounds of Examples Ex Synthetic # Structure/Name ^1H NMR/MS ($M + 1$) Method 1  (1S,2S)-N-(5-(1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 HCL salt. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 12.36 (br s, 1H), 11.16 (s, 1H), 8.76 (d, $J = 1.7$ Hz, 1H), 8.69-8.59 (m, 2H), 8.01 (s, 1H), 7.66 (t, $J = 2.8$ Hz, 1H), 7.27 (dd, $J = 2.0, 7.2$ Hz, 1H), 6.90 (s, 1H), 6.68 (dd, $J = 1.5, 3.1$ Hz, 1H), 5.06-4.81 (m, 1H), 2.17-2.13 (m, 1H), 1.73-1.59 (m, 1H), 1.17 (tdd, $J = 6.3, 9.0, 12.3$ Hz, 1H); LCMS (electrospray) m/z 336.00 ($M + H$) $^+$. A 2  (1S,2S)-N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 12.30-12.02 (m, 1H), 11.27-11.02 (m, 1H), 8.66-8.57 (m, 1H), 8.35-8.26 (m, 1H), 7.71 (d, $J = 1.1$ Hz, 1H), 7.69-7.65 (m, 1H), 6.96 (dd, $J = 2.0, 7.2$ Hz, 1H), 6.91 (s, 1H), 6.63-6.55 (m, 1H), 5.14-4.74 (m, 1H), 2.29-2.03 (m, 1H), 1.79-1.53 (m, 1H), 1.33-1.05 (m, 1H); LCMS (electrospray) m/z 370.1 ($M + H$) $^+$. A 3  (1S,2S)-2-fluoro-N-(5-(6-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2. ^1H NMR (400 MHz, methanol- d_4) δ 8.49 (d, $J = 7.2$ Hz, 1H), 8.38 (s, 1H), 7.59 (s, 1H), 7.57 (d, $J = 3.6$ Hz, 1H), 6.96 (s, 1H), 6.89 (dd, $J = 7.2, 2.0$ Hz, 1H), 6.75 (d, $J = 3.6$ Hz, 1H), 4.98-4.93 (m, 0.5H), 4.81-4.77 (m, 0.5H), 2.71 (s, 3H), 2.13-2.05 (m, 1H), 1.87-1.75 (m, 1H), 1.26-1.17 (m, 1H); LCMS (electrospray) m/z 350.1 ($M + H$) $^+$. A 4  (1S,2S)-N-(5-(6-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 TFA salt. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 11.98 (br s, 1H), 11.13 (s, 1H), 8.58 (d, $J = 7.2$ Hz, 1H), 8.11 (s, 1H), 7.65 (d, $J = 1.2$ Hz, 1H), 7.59 (dd, $J = 3.2, 2.4$ Hz, 1H), 6.93 (dd, $J = 7.2, 2.0$ Hz, 1H), 6.89 (s, 1H), 6.55 (dd, $J = 3.2, 2.0$ Hz, 1H), 5.05-4.82 (m, 1H), 2.17-2.10 (m, 1H), 1.73-1.61 (m, 1H), 1.21-1.13 (m, 1H); LCMS (electrospray) m/z 370.0 ($M + H$) $^+$. A 5  (1S,2S)-2-fluoro-N-(5-(4-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 12.14 (s, 1H), 11.13 (m, 1H), 8.62 (d, $J = 7.2$ Hz, 1H), 8.44 (d, $J = 10$ Hz, 1H), 7.82 (s, 1H), 7.59-7.58 (m, 1H), 7.08-7.06 (m, 1H), 6.91 (s, 1H), 6.62 (d, $J = 3.2$ Hz, 1H), 5.03-4.84 (m, 1H), 2.32-2.14 (m, 1H), 1.64-1.63 (m, 1H), 1.20-1.14 (m, 1H); LCMS (electrospray) m/z 354.1 ($M + H$) $^+$. A 6  (1S,2S)-2-fluoro-N-(5-(4-methoxy-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. ^1H NMR (400

MHz, DMSO-d.sub.6) δ 11.76 (br s, 1H), 11.07 (s, 1H), 8.50 (d, J = 7.2 Hz, 1H), 8.13 (s, 1H), 7.70-7.58 (m, 1H), 7.45-7.39 (m, 1H), 6.97 (dd, J = 7.2, 2.0 Hz, 1H), 6.84 (s, 1H), 6.82 (dd, J = 3.6, 2.0 Hz, 1H), 5.06-4.81 (m, 1H), 4.24 (s, 3H), 2.18-2.08 (m, 1H), 1.73-1.60 (m, 1H), 1.19-1.15 (m, 1H); LCMS (electrospray) m/z 366.2 (M + H)⁺. A 7  (1S,2S)-2-fluoro-N-(5-(3-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.65 (br s, 1H), 11.17-11.10 (m, 1H), 8.72 (d, J = 2.1 Hz, 1H), 8.63-8.58 (m, 1H), 8.39 (d, J = 2.1 Hz, 1H), 8.01 (d, J = 1.3 Hz, 1H), 7.54 (t, J = 2.4 Hz, 1H), 7.29 (dd, J = 2.1, 7.3 Hz, 1H), 6.93-6.86 (m, 1H), 5.05-4.82 (m, 1H), 2.17-2.10 (m, 1H), 1.72-1.61 (m, 1H), 1.22-1.14 (m, 1H); LCMS (electrospray) m/z 354.5 (M + H)⁺. A 8  (1S,2S)-2-fluoro-N-(5-(2-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.65 (s, 1H), 11.10 (s, 1H), 8.57 (d, J = 7.2 Hz, 1H), 8.53 (d, J = 2.0 Hz, 1H), 8.20 (d, J = 1.6 Hz, 1H), 7.92 (s, 1H), 7.21 (dd, J = 2.0, 7.2 Hz, 1H), 6.86 (s, 1H), 6.23 (s, 1H), 5.06-4.82 (m, 1H), 2.42 (s, 3H), 2.18-2.10 (m, 1H), 1.72-1.61 (m, 1H), 1.22-1.12 (m, 1H); LCMS (electrospray) m/z 350.3 (M + H)⁺. A 9  (1S,2S)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.89 (br s, 1H), 11.13 (s, 1H), 8.59 (d, J = 7.2 Hz, 1H), 8.21 (s, 1H), 7.62-7.59 (m, 1H), 7.57-7.52 (m, 1H), 6.91 (d, J = 2.0 Hz, 1H), 6.89 (s, 1H), 6.69-6.64 (m, 1H), 5.10-4.80 (m, 1H), 2.55 (s, 3H), 2.24-2.06 (m, 1H), 1.82-1.54 (m, 1H), 1.27-1.12 (m, 1H); LCMS (electrospray) m/z 350.1 (M + H)⁺. A 10  (1S,2S)-N-(5-(4-amino-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 13.95 (br s, 1H), 12.19 (s, 1H), 11.14 (s, 1H), 8.65 (d, J = 7.2 Hz, 1H), 8.00 (s, 1H), 7.67 (d, J = 0.8 Hz, 1H), 7.36-7.30 (m, 1H), 6.96 (dd, J = 3.2, 2.0 Hz, 1H), 6.92 (s, 1H), 6.84 (dd, J = 7.2, 2.0 Hz, 1H), 5.06-4.82 (m, 1H), 2.14 (td, J = 14.0, 7.2 Hz, 1H), 1.73-1.60 (m, 1H), 1.24-1.13 (m, 1H); LCMS (electrospray) m/z 351.3 (M + H)⁺. A 11  (1S,2S)-N-(5-(3-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.14 (s, 1H), 11.12 (s, 1H), 8.74 (d, J = 2.0 Hz, 1H), 8.60 (d, J = 7.2 Hz, 1H), 8.28 (d, J = 2.0 Hz, 1H), 8.03 (d, J = 1.2 Hz, 1H), 7.76 (d, J = 2.4 Hz, 1H), 7.29 (dd, J = 7.2, 2.0 Hz, 1H), 6.89 (s, 1H), 5.04-4.83 (m, 1H), 2.16-2.12 (m, 1H), 1.70-1.64 (m, 1H), 1.20-1.15 (m, 1H); LCMS (electrospray) m/z 370.1 (M + H)⁺. A 12  (1S,2S)-2-fluoro-N-(5-(3-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.49 (s, 1H), 11.12 (s, 1H), 8.64 (d, J = 2.0 Hz, 1H), 8.59 (d, J = 7.2 Hz, 1H), 8.35 (d, J = 2.0 Hz, 1H), 7.98 (d, J = 1.2 Hz, 1H), 7.31 (s, 1H), 7.28 (dd, J = 2.0, 7.2 Hz, 1H), 6.87 (s, 1H), 5.07-4.82 (m, 1H), 2.33 (s, 3H), 2.14 (td, J = 6.9, 13.9 Hz, 1H), 1.73-1.61 (m, 1H), 1.24-1.12 (m, 1H); LCMS (electrospray) m/z 350.1 (M + H)⁺. A 13  (1S,2S)-N-(5-(4-ethyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, methanol-d.sub.4) δ 8.50 (d, J = 7.2 Hz, 1H), 8.24 (s, 1H), 7.61 (d, J = 3.3 Hz, 1H), 7.57 (s, 1H), 6.96 (s, 1H), 6.92-6.75 (m, 2H), 5.32-4.93 (m, 1H), 3.16-3.06 (m, 2H), 2.08 (br d, J = 5.7 Hz, 1H), 1.91-1.67 (m, 1H), 1.37-1.04 (m, 4H); LCMS (electrospray) m/z 364.0 (M + H)⁺. A 14  (1S,2S)-2-fluoro-N-(5-(4-(methylthio)-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, methanol-d.sub.4) δ 8.46 (d, J = 7.2 Hz, 1H), 8.14 (s, 1H), 7.65-7.52 (m, 2H), 7.04-6.88 (m, 3H), 4.95 (dt, J = 3.6, 6.4 Hz, 1H), 4.79 (dt, J = 3.6, 6.4 Hz, 1H), 2.63 (s, 3H), 2.18-2.03 (m, 1H), 1.92-1.70 (m, 1H), 1.38-1.10 (m, 1H); LCMS (electrospray) m/z 382.0 (M + H)⁺. A 15  (1S,2S)-2-fluoro-N-(5-(4-isopropyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.91 (br s, 1H), 11.12 (s, 1H), 8.58 (d, J = 7.0 Hz, 1H), 8.12 (s, 1H), 7.57-7.54 (m, 1H), 7.53 (d, J = 0.9 Hz, 1H), 6.89 (s, 1H), 6.79-6.74 (m, 2H), 5.07-4.81 (m, 1H), 3.25 (spt, J =

7.0 Hz, 1H), 2.14 (td, J = 7.0, 13.8 Hz, 1H), 1.75-1.60 (m, 1H), 1.39 (d, J = 7.0 Hz, 6H), 1.17 (tdd, J = 6.3, 9.0, 12.3 Hz, 1H); LCMS (electrospray) m/z 378.1 (M + H)⁺. A 16  embedded image (1S,2S)-N-(5-(4-cyclopropyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.69 (br s, 1H), 11.11 (br s, 1H), 8.55 (br d, J = 6.3 Hz, 1H), 8.10 (br s, 1H), 7.62 (br s, 1H), 7.46 (br s, 1H), 6.94 (br d, J = 6.3 Hz, 1H), 6.87 (br s, 1H), 6.51 (br s, 1H), 5.12-4.76 (m, 1H), 2.24 (br s, 1H), 2.14 (br s, 1H), 1.78-1.56 (m, 1H), 1.17 (br s, 1H), 0.93 (br d, J = 5.8 Hz, 2H), 0.85 (br s, 2H); LCMS (electrospray) m/z 376.5 (M + H)⁺. A 17  embedded image (1S,2S)-N-(5-(4-cyano-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.51 (br s, 1H), 11.18 (s, 1H), 8.71 (d, J = 7.2 Hz,), 8.56 (s, 1H), 7.93-7.90 (m, 1H), 7.90-7.88 (m, 1H), 7.12 (dd, J = 7.2, 2.0 Hz, 1H), 6.96 (s, 1H), 6.71 (dd, J = 3.6, 2.0 Hz, 1H), 5.06-4.83 (m, 1H), 2.18-2.11 (m, 1H), 1.74-1.62 (m, 1H), 1.21-1.15 (m, 1H); LCMS (electrospray) m/z 361.0 (M + H)⁺. A 18  embedded image (1S,2S)-N-(5-(2-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide. 2 TFA salt .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.65 (br s, 1H), 11.11 (s, 1H), 8.65 (d, J = 2.4 Hz, 1H), 8.60 (d, J = 7.2 Hz, 1H), 8.29 (d, J = 2.4 Hz, 1H), 7.95 (s, 1H), 7.23 (dd, J = 7.2, 2.0 Hz, 1H), 6.88 (s, 1H), 6.58 (s, 1H), 5.05-4.84 (m, 1H), 2.16- 2.13 (m, 1H), 1.70-1.60 (m, 1H), 1.20-1.01 (m, 1H); LCMS (electrospray) m/z 370.0 (M + H)⁺. A 19  embedded image N-(5-(4-methyl-1H-pyrrolo[2,3- b]pyridin-5-yl)pyrazolo[1,5- alpyridin-2-yl)cyclopropanecarboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.69 (br s, 1H), 11.07 (s, 1H), 8.56 (d, J = 7.1 Hz, 1H), 8.13 (s, 1H), 7.56 (d, J = 1.0 Hz, 1H), 7.52- 7.46 (m, 1H), 6.87 (dd, J = 2.0, 7.1 Hz, 1H), 6.85 (s, 1H), 6.58 (dd, J = 1.8, 3.4 Hz, 1H), 2.49 (br s, 3H), 2.00-1.87 (m, 1H), 0.85-0.80 (m, 4H); LCMS (electrospray) m/z 332.2 (M + H)⁺. A 20  embedded image (1R,2R)-2-fluoro-N-(5-(4-methyl- 1H-pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.69 (br s, 1H), 11.11 (br s, 1H), 8.57 (d, J = 7.1 Hz, 1H), 8.14 (s, 1H), 7.61-7.55 (m, 1H), 7.49 (d, J = 3.4 Hz, 1H), 6.90-6.86 (m, 2H), 6.58 (d, J = 3.4 Hz, 1H), 5.08- 4.81 (m, 1H), 2.49 (s, 3H), 2.14 (td, J = 6.9, 14.0 Hz, 1H), 1.76- 1.58 (m, 1H), 1.17 (tdd, J = 6.2, 9.1, 12.3 Hz, 1H); LCMS (electrospray) m/z 350.1 (M + H)⁺. A 21  embedded image (1R,2S)-2-fluoro-N-(5-(4-methyl-1H- pyrrolo[2,3- b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2- yl)cyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.68 (br s, 1H), 11.23 (s, 1H), 8.57 (d, J = 7.1 Hz, 1H), 8.13 (s, 1H), 7.58 (d, J = 1.0 Hz, 1H), 7.53- 7.45 (m, 1H), 6.89 (dd, J = 2.0, 7.1 Hz, 1H), 6.82 (s, 1H), 6.58 (dd, J = 1.8, 3.4 Hz, 1H), 5.05-4.81 (m, 1H), 2.49 (br s, 3H), 2.44 (br d, J = 10.5 Hz, 1H), 1.61-1.45 (m, 1H), 1.26 (qd, J = 6.5, 13.1 Hz, 1H); LCMS (electrospray) m/z 350.0 (M + H)⁺. A 22  embedded image (1S,2R)-2-fluoro-N-(5-(4-methyl-1H- pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2- yl)cyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.68 (br s, 1H), 11.23 (br s, 1H), 8.57 (d, J = 7.1 Hz, 1H), 8.13 (s, 1H), 7.60-7.54 (m, 1H), 7.49 (d, J = 3.4 Hz, 1H), 6.89 (dd, J = 2.0, 7.2 Hz, 1H), 6.82 (s, 1H), 6.58 (d, J = 3.4 Hz, 1H), 5.06-4.81 (m, 1H), 2.49 (br s, 3H), 2.47-2.41 (m, 1H), 1.59-1.47 (m, 1H), 1.26 (qd, J = 6.5, 13.1 Hz, 1H); LCMS (electrospray) m/z 350.1 (M + H)⁺. A 23 0  embedded image N-(5-(4-chloro-1H-pyrrolo[2,3- b]pyridin-5-yl)pyrazolo[1,5- alpyridin-2- yl)cyclopropanecarboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.16 (br s, 1H), 11.08 (s, 1H), 8.60 (d, J = 7.1 Hz, 1H), 8.29 (s, 1H), 7.69 (d, J = 1.0 Hz, 1H), 7.67 (br d, J = 3.1 Hz, 1H), 6.95 (dd, J = 2.0, 7.1 Hz, 1H), 6.89 (s, 1H), 6.58 (d, J = 3.3 Hz, 1H), 2.00- 1.86 (m, 1H), 0.87-0.78 (m, 4H); LCMS (electrospray) m/z 352.1 (M + H)⁺. A 24  embedded image (1S,2S)-2-fluoro-N-(5-(3-fluoro-4- methyl-1H-pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2- yl)cyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.53 (br s, 1H), 11.12 (s, 1H), 8.58 (d, J = 7.3 Hz, 1H), 8.17 (s, 1H), 7.60 (s, 1H), 7.47 (s, 1H), 6.90-6.86 (m, 2H), 5.07-4.81 (m, 1H), 2.59 (s, 3H), 2.14 (br s, 1H), 1.79-1.55 (m, 1H), 1.28- 1.09 (m, 1H); LCMS (electrospray) m/z 368.2 (M + H)⁺. A 25  embedded image (1S,2S)-N-(5-(3-chloro-4-methyl- 1H-pyrrolo[2,3-b]pyridin-5-

yl)pyrazolo[1,5-a]pyridin-2-yl)-2- fluorocyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.98 (d, J = 10.9 Hz, 1H), 11.06 (s, 1H), 8.58 (d, J = 7.1 Hz, 1H), 8.15 (s, 1H), 7.71 (br d, J = 10.3 Hz, 1H), 7.61-7.56 (m, 1H), 6.88 (s, 1H), 6.87-6.84 (m, 1H), 5.08- 4.81 (m, 1H), 2.70 (s, 3H), 2.18- 2.09 (m, 1H), 1.75 (br d, J = 4.5 Hz, 1H), 1.23-1.10 (m, 1H); LCMS (electrospray) m/z 384.2 (M + H)⁺. A 26  (1S,2R)-N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2-yl)-2- fluorocyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.17 (br s, 1H), 11.26 (s, 1H), 8.62 (d, J = 7.2 Hz, 1H), 8.29 (s, 1H), 7.70 (d, J = 1.1 Hz, 1H), 7.69- 7.66 (m, 1H), 6.97 (dd, J = 2.0, 7.2 Hz, 1H), 6.87 (s, 1H), 6.60- 6.56 (m, 1H), 2.47-2.41 (m, 1H), 1.62-1.46 (m, 1H), 1.32-1.20 (m, 1H); LCMS (electrospray) m/z 370.1 (M + H)⁺. A 27  (1R,2S)-N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2-yl)-2- fluorocyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.17 (br s, 1H), 11.26 (s, 1H), 8.62 (d, J = 7.2 Hz, 1H), 8.29 (s, 1H), 7.70 (d, J = 1.0 Hz, 1H), 7.67 (br d, J = 3.1 Hz, 1H), 6.97 (dd, J = 2.0, 7.2 Hz, 1H), 6.87 (s, 1H), 6.58 (d, J = 3.3 Hz, 1H), 5.04- 4.80 (m, 1H), 2.45-2.41 (m, 1H), 1.61-1.47 (m, 1H), 1.27 (qd, J = 6.5, 13.0 Hz, 1H); LCMS (electrospray) m/z 370.1 (M + H)⁺. A 28  (1R,2R)-N-(5-(4-chloro-1H- pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2-yl)-2- fluorocyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.17 (br s, 1H), 11.14 (br s, 1H), 8.61 (br d, J = 6.6 Hz, 1H), 8.30 (s, 1H), 7.71 (br s, 1H), 7.67 (br d, J = 2.3 Hz, 1H), 6.99-6.93 (m, 1H), 6.91 (br s, 1H), 6.58 (br d, J= 1.7 Hz, 1H), 5.07-4.78 (m, 1H), 2.27-2.04 (m, 1H), 1.79-1.54 (m, 1H), 1.32-1.04 (m, 1H); LCMS (electrospray) m/z 370.0 (M + H)⁺. A 29  (1S,2S)-N-(5-(3,4-dichloro-1H- pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2-yl)-2- fluorocyclopropane-1-carboxamide .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.14 (s, 1H), 8.61 (d, J = 7.2 Hz, 1H), 8.33 (s, 1H), 7.86 (s, 1H), 7.70 (d, J = 1.1 Hz, 1H), 6.95 (dd, J = 2.0, 7.1 Hz, 1H), 6.92 (s, 1H), 5.13-4.75 (m, 1H), 2.18-2.10 (m, 1H), 2.07 (s, 1H), 1.73-1.60 (m, 1H), 1.24-1.11 (m, 1H); LCMS (electrospray) m/z 404.0 (M + H)⁺. A 30  (1S,2S)-N-(5-(4-chloro-3-fluoro-1H- pyrrolo[2,3-b]pyridin-5- yl)pyrazolo[1,5-a]pyridin-2-yl)-2- fluorocyclopropane-1-carboxamide 1H NMR (400 MHz, DMSO-d.sub.6) δ 12.02 (s, 1H), 11.14 (s, 1H), 8.62 (d, J = 7.0 Hz, 1H), 8.33 (s, 1H), 7.72 (d, J = 1.1 Hz, 1H), 7.68 (d, J= 2.0 Hz, 1H), 6.96 (dd, J = 2.0, 7.1 Hz, 1H), 6.92 (s, 1H), 5.07- 4.79 (m, 1H), 2.20-2.07 (m, 1H), 1.78-1.56 (m, 1H), 1.26-1.10 (m, 1H); LCMS (electrospray) m/z 388.0 (M + H)⁺. A

Evaluation of Compounds

RIP2 Kinase Assay

(94) GST-tagged recombinant human RIPK2 (25 ng) is incubated with 5 μ L of compounds (0.5% DMSO), 5 μ L of MBP (0.5 μ g/ μ L) and 5 μ L of ATP (25 UM) in buffer (40 mM Tris, 7.5; 20 mM MgCl.sub.2; 0.1 mg/ml BSA; 50 UM DTT). The assay was started by incubating the reaction mixture in a 96-well plate at 30° C. for 60 minutes. After the incubation, 25 μ L ADP-Glo reagent was added, and the reaction was incubated at 30° C. for 40 minutes to stop the reaction and degrade residual ATP. The ADP product was then converted to ATP by adding 50 μ L per well of detection reagent. Luminescence was detected after 30 minutes room temperature incubation with the Molecular device I3X plate reader. The IC.sub.50 values were calculated from a series of percent inhibition values determined at a range of inhibitor concentration using software routines as implemented in the GraphPad Prism 7 software or SigmaPlot 13.0.

(95) c-abl Kinase Assay

(96) ADP-Glo assay kit was purchased from Promega. Magnesium chloride (MgCl.sub.2), bovine serum albumin (BSA), ethylene glycol-bis(3-aminoethyl ether)-N,N,N',N'-tetraacetic acid (EGTA), tween-20, 1,4-dithiothreitol (DTT) and dimethyl sulfoxide (DMSO) were purchased from Sigma-Aldrich. HEPES buffer was purchased from Gibco. ABL1 kinase and Abltide were purchased from Signalchem.

(97) c-Abl kinase activity was measured by Promega's ADP-Glo™ Assay. In this assay, His-tagged

recombinant human ABL1 (0.25 ng/ μ L) is incubated with 5 μ L of compounds (0.5% DMSO), 5 μ L of Abltide (0.01 μ g/ μ L) and 5 μ L of ATP (25 μ M) in buffer (50 mM HEPES, 7.5; 10 mM MgCl₂; 1 mM EGTA; 0.05% BSA; 0.01% Tween-20; 2 mM DTT.). The assay was started by incubating the reaction mixture in a 96-well plate at 30° C. for 30-min. After the incubation, 25 μ L ADP-Glo reagent was added and the reaction was incubated at room temperature for 40-min to stop the reaction and degrade residual ATP. The ADP product was then converted to ATP by adding 50 μ L per well of detection reagent. Luminescence was detected after 30-min room temperature incubation with the Molecular device I3X plate reader. The IC₅₀ values were calculated from a series of percent inhibition values determined at a range of inhibitor concentration using software routines as implemented in the GraphPad Prism 7 software and Sigma Plot 13.0.

(98) LRRK2 and LRRK2 G2019S Kinase Assay

(99) The LRRK2 and LRRK2 G2019S kinase assays were performed using the Adapta™ technology in ThermoFisher Scientific. This experiment was carried out according to the supplier's protocol. Briefly, assay conditions were as follows. The mixture of substrate (LRRKtide) and each kinase was prepared in 50 mM Tris pH 8.5, 0.01% BRIJ-35, 10 mM MgCl₂, 1 mM EGTA, 0.02% NaN₃. Assays were performed in the presence of 70 μ M ATP and 100 μ M ATP (KmATP) in LRRK2 and LRRK2 G2019S, respectively. The reaction was progressed at room temperature for 1 hour. Upon completion of kinase reaction, 5 μ L of Detection Mix was added and after 60 min incubation time, the emission ratio of 665/615 nm was calculated.

(100) IL-6 Secretion in BV-2 Cell

(101) BV-2 mouse microglia cells were kindly provided from Dr. Bae in Korea Institute of Science and Technology (KIST). BV-2 cells were thawed and suspended in Dulbecco's modified eagle medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% Penicillin/Streptomycin. Cells were seeded into 96 well plates at 2 $\times$ 10⁴ cells/well and allowed to attach for 24 hr. On the day of the experiment, various concentration of compounds were treated, and cells were stimulated with 10 μ g/ml L18-MDP. After 24 hr incubation, supernatant was collected for cytokine assay.

(102) Cytokine secretion was measured after 24 hr post stimulation using IL-6 ELISA kit (R&D system) as suggested by the manufacturer. Absorbance at 450 nm was measured using SpectraMax I3X microplate reader (Molecular Device). Values of media-only wells were subtracted and % inhibition for each compound concentration relative to the DMSO/L18-MDP-treated controls (100%) was calculated. The IC₅₀ values were calculated from a series of percent activity values using software as implemented in the GraphPad Prism 7 software or SigmaPlot 13.0.

(103) Table 2 shows IC₅₀ values for the compounds of Examples.

(104) TABLE-US-00002 TABLE 2 In vitro kinase activity and IL-6 secretion in BV-2 cell. RIP2 c-Abl LRRK2 LRRK2_ IL-6 IC₅₀ IC₅₀ (WT) G2019S IC₅₀ Example (nM) (nM) IC₅₀ (nM) IC₅₀ (nM) (nM) 1 11.04 0.71 55.7 28.0 187.2 2 6.67 7.6 36.5 15.0 68.58 3 25.02 0.7 Not tested Not tested 331.94 4 13.95 1.2 Not tested Not tested 350.6 5 48.37 11.2 Not tested Not tested 600.11 6 168.4 9.0 Not tested Not tested 805.89 7 24.26 3.6 Not tested Not tested 732.22 8 84.27 4.0 Not tested Not tested 624.89 9 8.77 3.1 Not tested Not tested 100.85 10 91.51 54.2 Not tested Not tested 3943.46 11 33.37 3.2 Not tested Not tested 654.14 12 88.64 2.5 Not tested Not tested 686.77 13 128.6 12.3 Not tested Not tested Not tested 14 195.4 12.2 Not tested Not tested Not tested 15 828.2 158.7 Not tested Not tested Not tested 16 204.4 13.0 Not tested Not tested Not tested 17 70.2 20.2 Not tested Not tested Not tested 18 30.51 3.6 Not tested Not tested Not tested 19 11.4 2.4 Not tested Not tested 82.89 20 8.37 3.1 Not tested Not tested 65.54 21 8.25 3.3 Not tested Not tested 61.38 22 15.88 4.2 Not tested Not tested 57.08 23 19.88 9.2 Not tested Not tested Not tested 24 14.87 2.9 Not tested Not tested 75.74 25 32.21 1.7 Not tested Not tested Not tested 26 16.06 19.4 Not tested Not tested Not tested 27 15.12 12.4 Not tested Not tested 162.15 28 21.48 13.9 Not tested Not tested Not tested 29 15.4 4.3 Not tested Not tested 176.76 30 10.2 7.2 Not tested Not tested 107.07

Claims

1. A compound of Formula (I): ##STR00038## or a pharmaceutically acceptable salt thereof, wherein: R^{sup.1} is cyclopropyl, cyclobutyl, 3- or 4-membered heterocyclyl, or —CF_{sub.3}, wherein R^{sup.1} is optionally substituted with one or more groups selected from the group consisting of halo, alkyl, hydroxyalkyl, haloalkyl, and monoalkylaminoalkyl, R^{sup.2} and R^{sup.3} are independently —H, halo, C_{sub.1}-C_{sub.3} alkyl, alkoxy, —CF_{sub.3}, or —OCF_{sub.3}, R^{sup.4}, R^{sup.5}, R^{sup.6} and R^{sup.7} are independently —H, halo, C_{sub.1}-C_{sub.3} alkyl, alkoxy, aryl, heteroaryl, cycloalkyl, heterocyclyl, heteroalkyl, amino, —SMe, or —CN, wherein R^{sup.4}, R^{sup.5}, R^{sup.6} and R^{sup.7} are optionally substituted with one or more groups selected from the group consisting of halo, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, hydroxyalkyl, trimethylsilylethoxymethyl, —NO_{sub.2}, —NR_{sub.a}R_{sub.b}, —NR_{sub.a}C(=O)R_{sub.b}, —NR_{sub.a}C(=O)NR_{sub.a}R_{sub.b}, —NR_{sub.a}C(=O)OR_{sub.b}, —OR_{sub.a}, —CN, —C(=O)R_{sub.a}, —C(=O)OR_{sub.a}, —C(=O)NR_{sub.a}R_{sub.b}, —OC(=O)R_{sub.a}, —OC(=O)OR_{sub.a}, and —OC(=O)NR_{sub.a}R_{sub.b}, and R_{sub.a} and R_{sub.b} are independently —H, halo, amino, alkyl, or haloalkyl.
2. The compound of claim 1 or a pharmaceutically acceptable salt thereof, which is a compound of Formula (II): ##STR00039## wherein R^{sup.8} is selected from the group consisting of H, halo, C_{sub.1}-C_{sub.3} alkyl, C_{sub.1}-C_{sub.3} hydroxyalkyl, and C_{sub.1}-C_{sub.3} haloalkyl.
3. The compound of claim 2 or a pharmaceutically acceptable salt thereof, wherein R^{sup.2} and R^{sup.3} are —H; R^{sup.4}, R^{sup.5}, R^{sup.6} and R^{sup.7} are independently —H, halo, C_{sub.1}-C_{sub.3} alkyl, alkoxy, aryl, heteroaryl, cycloalkyl, heterocyclyl, heteroalkyl, amino, —SMe, or —CN, wherein R^{sup.4}, R^{sup.5}, R^{sup.6} and R^{sup.7} are optionally substituted with one or more groups selected from the group consisting of halo, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, hydroxyalkyl, trimethylsilylethoxymethyl, —NO_{sub.2}, —NR_{sub.a}R_{sub.b}, —NR_{sub.a}C(=O)R_{sub.b}, —NR_{sub.a}C(=O)NR_{sub.a}R_{sub.b}, —NR_{sub.a}C(=O)OR_{sub.b}, —OR_{sub.a}, —CN, —C(=O)R_{sub.a}, —C(=O)OR_{sub.a}, —C(=O)NR_{sub.a}R_{sub.b}, —OC(=O)R_{sub.a}, —OC(=O)OR_{sub.a}, and —OC(=O)NR_{sub.a}R_{sub.b}; and R_{sub.a} and R_{sub.b} are independently —H, halo, amino, alkyl, or haloalkyl.
4. The compound of claim 2 or a pharmaceutically acceptable salt thereof, wherein the compound is selected from the group consisting of: N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropanecarboxamide; (1R,2R)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1R,2S)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; (1S,2R)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl) cyclopropane-1-carboxamide; N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropanecarboxamide; (1S,2R)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1R,2S)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; and (1R,2R)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide.
5. The compound of claim 1 or a pharmaceutically acceptable salt thereof, which is a compound of Formula (III): ##STR00040## wherein R^{sup.8} is selected from the group consisting of —H, halo, C_{sub.1}-C_{sub.3} alkyl, C_{sub.1}-C_{sub.3} hydroxyalkyl, and C_{sub.1}-C_{sub.3} haloalkyl.
6. The compound of claim 5 or a pharmaceutically acceptable salt thereof, wherein R^{sup.2} and R^{sup.3} are —H; R^{sup.4}, R^{sup.5}, R^{sup.6} and R^{sup.7} are independently —H, halo, C_{sub.1}-C_{sub.3} alkyl, alkoxy, aryl, heteroaryl, cycloalkyl, heterocyclyl, heteroalkyl, amino, —SMe, or —CN, wherein R^{sup.4}, R^{sup.5}, R^{sup.6} and R^{sup.7} are optionally substituted with one or more groups selected from the group consisting of halo, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl,

hydroxyalkyl, trimethylsilylethoxymethyl, —NO.sub.2, —NR.sub.aR.sub.b, —NR.sub.aC(=O)R.sub.b, —NR.sub.aC(=O)NR.sub.aR.sub.b, —NR.sub.aC(=O)OR.sub.b, —OR.sub.a, —CN, —C(=O)R.sub.a, —C(=O)OR.sub.a, —C(=O)NR.sub.aR.sub.b, —OC(=O)R.sub.a, —OC(=O)OR.sub.a, and —OC(=O)NR.sub.aR.sub.b; R.sub.a and R.sub.b are independently —H, halo, amino, alkyl, or haloalkyl.

7. The compound of claim 5 or a pharmaceutically acceptable salt thereof, wherein the compound is selected from the group consisting of: (1S,2S)—N-(5-(1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(6-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)—N-(5-(6-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-methoxy-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(3-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(2-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-amino-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(3-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(3-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-ethyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-(methylthio)-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(4-isopropyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-cyclopropyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(4-cyano-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(2-chloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)-2-fluoro-N-(5-(3-fluoro-4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)cyclopropane-1-carboxamide; (1S,2S)—N-(5-(3-chloro-4-methyl-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; (1S,2S)—N-(5-(3,4-dichloro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide; and (1S,2S)—N-(5-(4-chloro-3-fluoro-1H-pyrrolo[2,3-b]pyridin-5-yl)pyrazolo[1,5-a]pyridin-2-yl)-2-fluorocyclopropane-1-carboxamide.

8. The compound of claim 1, wherein the salt is hydrochloric acid salt, tartaric acid salt, phosphoric acid salt, or maleic acid salt.

9. A pharmaceutical composition comprising a therapeutically effective amount of the compound of claim 1 or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable carrier.

10. The pharmaceutical composition of claim 9, further comprising one or more active ingredients that are useful to treat a neurodegenerative disease.

11. A method for treating an inflammatory and autoimmune disease in a subject, comprising: administering to the subject in need thereof a therapeutically effective amount of the compound of claim 1 or a pharmaceutically acceptable salt thereof.

12. The method of claim 11, wherein the inflammatory and autoimmune is a neuroinflammation disease.

13. The method of claim 11, wherein the neuroinflammation disease is α -synucleinopathy, Parkinson's disease, dementia with Lewy body, multiple system atrophy (MSA), Alzheimer's

disease, or amyotrophic lateral sclerosis (ALS).

14. The method of claim 11, wherein the subject is a human.
